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Sample Solutions 05

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1 Warm-Up

We use the following Lemma in the script:

Lemma 1. *If c is drawn uniformly at random from X, then $\mathbb{E}[\phi(X, c)] \leq 2\,\phi(X, \mu_X)$.*

Let $C^* = \{c_1^*, \ldots, c_k^*\}$ be the optimal unrestricted centers, and let $A_1^*, \ldots, A_k^*$ be the corresponding optimal clusters. The optimal center c_i^* is the centroid of A_i^* (i.e., $c_i^* = \mu_{A_i^*}$), and the optimal cost is $\phi^* = \sum_{i=1}^k \phi(A_i^*, \mu_{A_i^*})$.

Consider a random restricted set $C_{rand} = \{c_1, \ldots, c_k\}$ formed by choosing each c_i independently and uniformly at random from the points in A_i^*.

We can bound the expected cost of this random set by assigning all points in A_i^* to their corresponding random center c_i:

$$\mathbb{E}[\phi(X, C_{rand})] \leq \mathbb{E}\left[\sum_{i=1}^k \phi(A_i^*, c_i)\right] = \sum_{i=1}^k \mathbb{E}[\phi(A_i^*, c_i)].$$

By the Lemma above, the expected cost for a single cluster, where the center c_i is chosen uniformly at random from A_i^*, is at most twice the optimal cost for that cluster:

$$\mathbb{E}[\phi(A_i^*, c_i)] \leq 2 \cdot \phi(A_i^*, \mu_{A_i^*}).$$

Substituting this into the sum gives:

$$\mathbb{E}[\phi(X, C_{rand})] \leq \sum_{i=1}^k 2 \cdot \phi(A_i^*, \mu_{A_i^*}) = 2 \sum_{i=1}^k \phi(A_i^*, \mu_{A_i^*}) = 2\phi^*.$$

Since the expected cost over all such random sets $C_{rand} \subseteq X$ is at most $2\phi^*$, there must exist at least one specific set $C_{exist} \subseteq X$ whose cost is at most the average. Therefore,

$$\hat{\phi}^* = \min_{C \subseteq X, |C| = k} \phi(X, C) \leq \phi(X, C_{exist}) \leq 2\phi^*.$$

2 Analyzing the Final Step of k-means||

Exercise 2A

Let $c \in C$ be an arbitrary center. We apply the squared triangle inequality with $u = x - \pi(x)$ and $v = \pi(x) - c$:

$$\begin{aligned} \|x - c\|^2 &= \|(x - \pi(x)) + (\pi(x) - c)\|^2 \\ &\leq 2\|x - \pi(x)\|^2 + 2\|\pi(x) - c\|^2. \end{aligned}$$

This inequality holds for all $c \in C$. Since the left side of our target inequality is a minimum, we take the $\min_{c \in C}$ of both sides. The term $2\|x - \pi(x)\|^2$ is constant with respect to c, so it can be pulled out of the minimum on the right.

$$\begin{aligned}\min_{c \in C} \|x - c\|^2 &\leq \min_{c \in C} \left(2\|x - \pi(x)\|^2 + 2\|\pi(x) - c\|^2\right) \\ &\leq 2\|x - \pi(x)\|^2 + \min_{c \in C} \left(2\|\pi(x) - c\|^2\right) \\ &\leq 2\|x - \pi(x)\|^2 + 2\min_{c \in C} \|\pi(x) - c\|^2.\end{aligned}$$

Exercise 2B

We start with the definition of $\phi(X, C)$ and apply the result from Exercise 2.A to each term in the sum.

$$\begin{aligned}\phi(X, C) &= \sum_{x \in X} \min_{c \in C} \|x - c\|^2 \\ &\leq \sum_{x \in X} \left(2\|x - \pi(x)\|^2 + 2\min_{c \in C} \|\pi(x) - c\|^2\right) && \text{(by 2.A)} \\ &= 2\sum_{x \in X} \|x - \pi(x)\|^2 + 2\sum_{x \in X} \min_{c \in C} \|\pi(x) - c\|^2.\end{aligned}$$

We analyze the two terms separately:

1. **First Term:** By definition, $\pi(x)$ is the closest point in B to x. Thus, $\|x - \pi(x)\|^2 = \min_{b \in B} \|x - b\|^2$. The sum $\sum_{x \in X} \min_{b \in B} \|x - b\|^2$ is the definition of $\phi(X, B)$. Therefore, $2\sum_{x \in X} \|x - \pi(x)\|^2 = 2\phi(X, B)$.

2. **Second Term:** We group the sum by the points $b \in B$ that serve as the closest center $\pi(x)$ for each x.

$$\begin{aligned}\sum_{x \in X} \min_{c \in C} \|\pi(x) - c\|^2 &= \sum_{b \in B} \sum_{x \in X : \pi(x) = b} \min_{c \in C} \|\pi(x) - c\|^2 \\ &= \sum_{b \in B} \sum_{x \in X : \pi(x) = b} \min_{c \in C} \|b - c\|^2 && \text{(since } \pi(x) = b\text{)} \\ &= \sum_{b \in B} \left(\left|\{x \in X : \pi(x) = b\}\right| \cdot \min_{c \in C} \|b - c\|^2\right) \\ &= \sum_{b \in B} w_b \cdot \min_{c \in C} \|b - c\|^2 && \text{(by def. of } w_b\text{)} \\ &= \phi_w(B, C). && \text{(by def. of } \phi_w\text{)}\end{aligned}$$

Substituting these two results back into the inequality gives:

$$\phi(X, C) \leq 2\phi(X, B) + 2\phi_w(B, C).$$

Exercise 2C

We begin with the result from Exercise 2B:

$$\phi(X, C) \leq 2\phi(X, B) + 2\phi_w(B, C).$$

Conditioning on B and taking expectations gives

$$\mathbb{E}[\phi(X, C) \mid B] \leq 2\phi(X, B) + 2\mathbb{E}[\phi_w(B, C) \mid B].$$

From assumption (A2), we know that the set C returned by weighted k-means++ satisfies

$$\mathbb{E}[\phi_w(B, C) \mid B] \leq \beta \cdot \min_{\substack{C' \subseteq \mathbb{R}^d \\ |C'|=k}} \phi_w(B, C').$$

Substituting this into our first inequality, we get

$$\mathbb{E}[\phi(X, C) \mid B] \leq 2\phi(X, B) + 2\beta \cdot \min_{\substack{C' \subseteq \mathbb{R}^d \\ |C'|=k}} \phi_w(B, C').$$

Now, consider the set C^*, the optimal k centers for the original dataset X. C^* is a set of size k. The minimum cost over *all* possible sets C' of size k must be less than or equal to the cost of this *specific* set C^*. Therefore,

$$\min_{\substack{C' \subseteq \mathbb{R}^d \\ |C'|=k}} \phi_w(B, C') \leq \phi_w(B, C^*).$$

Substituting this final inequality completes the proof:

$$\mathbb{E}[\phi(X, C) \mid B] \leq 2\phi(X, B) + 2\beta \cdot \phi_w(B, C^*).$$

Exercise 2D

We start from the definition of $\phi_w(B, C^*)$ and expand the weights w_b:

$$\begin{aligned}
\phi_w(B, C^*) &= \sum_{b \in B} w_b \cdot \min_{c \in C^*} \|b - c\|^2 \\
&= \sum_{b \in B} \left(\left|\{x \in X : \pi(x) = b\}\right|\right) \cdot \min_{c \in C^*} \|b - c\|^2 \\
&= \sum_{b \in B} \sum_{x \in X : \pi(x) = b} \min_{c \in C^*} \|b - c\|^2 \\
&= \sum_{x \in X} \min_{c \in C^*} \|\pi(x) - c\|^2. && \text{(since } b = \pi(x)\text{)}
\end{aligned}$$

This gives the reformulation suggested in the hint. Now, for any $x \in X$ and $c \in C^*$, we apply the squared triangle inequality with $u = \pi(x) - x$ and $v = x - c$:

$$\|\pi(x) - c\|^2 = \|(\pi(x) - x) + (x - c)\|^2 \leq 2\|\pi(x) - x\|^2 + 2\|x - c\|^2.$$

This holds for all $c \in C^*$. We take the minimum over $c \in C^*$ on both sides:

$$\begin{aligned}
\min_{c \in C^*} \|\pi(x) - c\|^2 &\leq \min_{c \in C^*} \left(2\|\pi(x) - x\|^2 + 2\|x - c\|^2\right) \\
&\leq 2\|\pi(x) - x\|^2 + 2 \min_{c \in C^*} \|x - c\|^2.
\end{aligned}$$

Now, we sum this inequality over all $x \in X$:

$$\sum_{x \in X} \min_{c \in C^*} \|\pi(x) - c\|^2 \leq \sum_{x \in X} \left(2\|\pi(x) - x\|^2 + 2 \min_{c \in C^*} \|x - c\|^2\right)$$

$$\sum_{x \in X} \min_{c \in C^*} \|\pi(x) - c\|^2 \leq 2 \sum_{x \in X} \|\pi(x) - x\|^2 + 2 \sum_{x \in X} \min_{c \in C^*} \|x - c\|^2.$$

We identify each term:

- **Left side:** $\phi_w(B, C^*)$ (from our first step).
- **First term on RHS:** $2 \sum_{x \in X} \|x - \pi(x)\|^2 = 2\phi(X, B)$.
- **Second term on RHS:** $2 \sum_{x \in X} \min_{c \in C^*} \|x - c\|^2 = 2\phi(X, C^*) = 2\phi^*$.

Substituting these back gives the final inequality:

$$\phi_w(B, C^*) \leq 2\phi(X, B) + 2\phi^*.$$

Exercise 2E

We create a chain of inequalities by substituting the results from the previous exercises and the given assumptions. Taking expectations in the bound from Exercise 2C and using the law of total expectation, we obtain

$$\begin{aligned}\mathbb{E}[\phi(X,C)] &\le 2\mathbb{E}[\phi(X,B)] + 2\beta \cdot \mathbb{E}[\phi_w(B,C^*)] && \text{(by 2C)}\\ &\le 2\mathbb{E}[\phi(X,B)] + 2\beta \cdot (2\mathbb{E}[\phi(X,B)] + 2\phi^*) && \text{(by 2D)}\\ &= (2+4\beta)\mathbb{E}[\phi(X,B)] + 4\beta\phi^*.\end{aligned}$$

Now, we use assumption (A1), which states $\mathbb{E}[\phi(X,B)] \le \alpha \cdot \phi^*$:

$$\begin{aligned}\mathbb{E}[\phi(X,C)] &\le (2+4\beta)(\alpha \cdot \phi^*) + 4\beta\phi^*\\ &= (2\alpha + 4\alpha\beta + 4\beta)\phi^*.\end{aligned}$$

This proves the inequality in expectation, with the function $f(\alpha,\beta)$ defined as

$$f(\alpha,\beta) = 2\alpha + 4\alpha\beta + 4\beta.$$

3 Settled Clusters and k-means||

Exercise 3A

If $\phi(A^*, C) = 0$, the claim is immediate. If $\phi(A^*, \mu_{A^*}) = 0$, all points in A^* coincide, so adding any such point as a center makes the cluster settled. Thus, $A' = A^*$, and the claim follows. Otherwise, let $c \in A^*$ be drawn by D^2-sampling w.r.t. C, i.e., $\Pr[c = x] = \frac{\phi(x,C)}{\phi(A^*,C)}$ for $x \in A^*$. By the given lemma,

$$\mathbb{E}\big[\phi(A^*, C \cup \{c\})\big] \le 8\,\phi(A^*, \mu_{A^*}).$$

By Markov's inequality,

$$\Pr\big[\phi(A^*, C \cup \{c\}) \le 10\,\phi(A^*, \mu_{A^*})\big] \ge 1 - \frac{8}{10} = \frac{1}{5}.$$

By the definition of A', this means $\Pr[c \in A'] \ge 1/5$. Since c is sampled with probability proportional to $\phi(\cdot, C)$ on A^*, we obtain

$$\frac{\phi(A', C)}{\phi(A^*, C)} = \sum_{x \in A'} \Pr[c = x] = \Pr[c \in A'] \ge \frac{1}{5}.$$

Therefore,

$$\phi(A', C) \ge \tfrac{1}{5}\,\phi(A^*, C).$$

Exercise 3B

Recall that sampling any point from A' makes the cluster settled. Thus, the probability that A^* remains unsettled is at most the probability that no point of A' is sampled.

In one overseeding round, each $x \in X$ is independently included with probability $p(x) = \min\{1, \ell\,\phi(x,C)/\phi(X,C)\}$. If some $a \in A'$ has $p(a) = 1$, the cluster becomes settled with probability 1, so the bound is trivial.

Otherwise, for all $x \in A'$ we have $p(x) = \ell\,\phi(x,C)/\phi(X,C)$, and hence the probability that no point of A' is sampled is

$$\prod_{x \in A'} (1 - p(x)) \le \exp\Big(-\sum_{x \in A'} p(x)\Big) \le \exp\left(-\frac{\ell\,\phi(A^*, C)}{5\phi(X, C)}\right).$$

Here, the last inequality follows from Exercise 3A. Thus, A^* fails to be settled with probability at most the stated bound.